

Gabriel Pedde Ungureanu | Physics PhD | Quantitative Researcher

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Profile

PhD physicist (6 papers, ~60 citations) with a mathematics degree (110/110 *cum laude*): probability, stochastic modelling, numerical PDEs. I turn mathematical structure into fast, validated code — Gaussian-process inference from first principles, nonlinear PDE solvers, reinforcement learning at EuroHPC scale.

Research & Engineering Experience

SISSA

Trieste, Italy

Doctoral Researcher in Theoretical Physics

2022–2026

- Exact, non-perturbative results in quantum field theory via large-scale symbolic computation — generating functions (Hilbert series), combinatorial dualization algorithms — on SISSA's Ulysses cluster; **6 published papers**, ~60 citations.
- Built an automated “**AI co-scientist**” research pipeline: LLM hypothesis generation refereed by a **deterministic symbolic/numerical evaluator** (the model never grades itself); recovers exact General Relativity solutions to **machine precision** through self-improving evolutionary search.

Education

SISSA / ICTP Joint Programme

Trieste, Italy

Master in High Performance Computing and AI (MHPC)

2025–2026

Concurrent with the last year of PhD. Numerical analysis for PDEs, Bayesian statistics, machine learning, deep learning, reinforcement learning, and distributed GPU computing.

Università Cattolica del Sacro Cuore

Brescia, Italy

BSc Mathematics & MSc Physics (Laurea Triennale & Laurea Magistrale)

2017–2022

Real & functional analysis, probability, PDEs; theoretical physics, QFT, statistical mechanics. MSc thesis in **quantum complexity theory**, published in Physical Review D (28 citations).

Awards & Publications

6 peer-reviewed papers in theoretical physics (~60 citations) — [INSPIRE-HEP](#) | [Google Scholar](#)

Best graduate of the year, MSc Physics — Università Cattolica del Sacro Cuore

110/110 *cum laude* — MSc Physics | 110/110 *cum laude* — BSc Mathematics

Istituto Toniolo merit fellowship — BSc, Università Cattolica del Sacro Cuore

Selected Projects — Quantitative Modelling

Bayesian time series: **Quasi-periodic Gaussian Process built from scratch** (no GP libraries) to extract stellar rotation periods from noisy NASA Kepler light curves: Cholesky-factorised **log-marginal-likelihood maximisation**, Lomb-Scargle initialisation, full posterior predictive mean and uncertainty bands | [GitHub](#)

Nonlinear PDEs: Solved a stiff **nonlinear parabolic PDE** (Wetterich renormalization-group flow) two independent ways: method-of-lines **finite differences** (adaptive RK45) and a **physics-informed neural network** (PyTorch Lightning); located its fixed point via a **shooting method**, recovering critical exponent $\nu = 0.649$ vs 0.630 (Monte Carlo). | [GitHub](#)

Reinforcement learning: Implemented **RLOO/PPO from scratch** to fine-tune Llama-3.2-1B on GSM8K math reasoning with a rule-based reward and KL regularisation; **+11 pp answer accuracy** (35% → 46%). Distributed on **Leonardo** (CINECA). Stack: PyTorch, TRL, SLURM | [GitHub](#)

Performance engineering: CUDA kernel development through to cuBLAS; distributed dense linear algebra with **MPI + CUDA**. Lattice Boltzmann CFD solver: ~120× **GPU speedup** over CPU, 806 GB/s sustained bandwidth | [GitHub](#)

Technical Skills

Quant methods: Probability & stochastic processes, Bayesian inference, Gaussian processes, time series, Monte Carlo, numerical PDEs (FD/FEM), iterative solvers, optimisation

ML / AI: PyTorch, scikit-learn, reinforcement learning (REINFORCE, RLOO, PPO), CNNs, autoencoders, dimensionality reduction, clustering, LLM agents (LangGraph, RAG)

Programming: Python (NumPy, SciPy, pandas), C++, C, Fortran, Mathematica, Git, $\LaTeX$

Scale / HPC: CUDA, cuBLAS, MPI, OpenMP, SLURM, PETSc/SLEPc, Docker/Apptainer; Leonardo (CINECA), Ulysses (SISSA)

Languages: Italian (native) | English (C1) | Romanian (A2)

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